# C-STGB: Real-Time Financial Surveillance Architecture & Conformal Triage Platform

## (a) End-to-End Streaming Ingestion & Deep Spatial-Temporal Graph Neural Architecture

Real-Time Fast-Path Latency: 0.45 ms (P99: 0.82 ms) | 31,200 tx/s Throughput

### Stage 1: Streaming Ingest
Wire & Zero-Copy Memory

**SWIFT MT103 | ISO-20022 | Kafka**

UTXO Blockchains (Bitcoin, Eth)

**DuckDB Streaming Micro-Batches**

Apache Arrow Zero-Copy Buffer

**Causal Split:** $t_e \leq t_{\text{eval}}$ **(No Leak)**

**Ingestion Latency SLA: 0.11 ms**

### Stage 2: 12-D Invariants
Physical Flow & Hawkes

**Mass Flow:** $\Phi_{\text{flow}} = \ln(1 + \frac{A_{\text{out}}}{A_{\text{in}} + \varepsilon})$

Pass-Through Layering Detector

**Hawkes Arrival:** $\lambda_u(t) = \mu_u + \sum \alpha e^{-\beta \Delta t}$

Burst Acceleration Metric $(\mu_A, \sigma_A^2)$

**Directed Causal Taint:** $s_{\text{fwd}}, s_{\text{bwd}}$

**Tensor Descriptor:** $\mathbf{z}_{\text{inv}} \in \mathbb{R}^{12}$

### Stage 3: Temporal GNN
Tri-Band & Edge Gating

**3-Band Harmonic Temporal Attention**

$\tau_1 = 2.4\,\text{h}$ (burst), $\tau_2 = 24\,\text{h}$, $\tau_3 = 168\,\text{h}$

**Edge Gate:** $\hat{g}_{ij} = \sigma(\mathbf{w}_g^\top [\mathbf{h}_i \,\|\, \mathbf{h}_j \,\|\, \mathbf{e}_{ij}])$

Prunes 65.9% Chaff Camouflage Links

**Typology GraphSMOTE on Clusters** $C_k$

**GNN Forward SLA: 0.16 ms**

### Stage 4: Fusion & Risk
Adaptive Head & Bayes Tau*

**Adaptive Gate:** $\alpha_u = \sigma(\mathbf{w}_\alpha^\top [\mathbf{h}_u \,\|\, \mathbf{z}_{\text{inv}}])$

Unified Space: $\mathbf{h}_u^* = \alpha_u \mathbf{h}_u + (1 - \alpha_u)\mathbf{z}_{\text{inv}}$

**Cold-Start: 89.60% F1 for** $\deg(u) \leq 2$

**Cost-Sensitive Head:** $\tau^* = 0.18$

15:1 Asymmetric Ratio $(C_{\text{FN}}/C_{\text{FP}})$

**Minority Risk Output:** $\hat{p}_u \in [0, 1]$

## (b) Class-Conditional Conformal Risk Control (CRC) Triage Gate & 3-Tier Operational Dispatch

Finite-Sample Risk Bound: $P(Y \in \Gamma(X) \mid Y = y) \geq 99.0\%$ | 0.00% Empty Sets

### Holdout Calibration
Holdout Split: $\mathcal{D}_{\text{cal}}$

$n_0 = 3,306, \;\; n_1 = 372$

Non-Conformity Scores:

$s_i^{(1)} = 1 - \hat{p}_i$

$s_i^{(0)} = \hat{p}_i$

**Target Coverage: 99.0%**

**Class-Cond.**
**CRC Triage**
$1 - \alpha \geq 99\%$
$\hat{q}^{(0)}, \hat{q}^{(1)}$

### Tier 1: High-Risk Immediate Quarantine $\Gamma(u) = \{1\}$ (Decisive Illicit)
**18 alerts / 10,000 tx (0.18%)**

**Immediate Quarantine Hold | Automated Wire Freeze**

Generates FinCEN Form 111 SAR Dossier with GNNExplainer Attribution Subgraph & SHA-256 Merkle Audit Receipt

### Tier 2: Compliance Officer Review Desk $\Gamma(u) = \{0, 1\}$ (Abstention Boundary)
**49 alerts / 10,000 tx (0.49%) | Workload Reduction: 99.51%**

**Human-in-the-Loop 4-Eyes Compliance Review Queue**

Ambiguous transactions routed to senior examiners; cuts false alarms by 85.7%--95.8% vs rules; feeds Active Learning loop

### Tier 3: Automated Straight-Through Clearing $\Gamma(u) = \{0\}$ (Decisive Benign)
**9,933 transactions / 10,000 tx (99.33%)**

**Automated Wire-Speed Straight-Through Processing (STP)**

Instantaneous clearance under 0.45 ms SLA with mathematically bounded selective risk (residual illicit leakage < 0.13%)

## REAL-TIME PRODUCTION SLAS | STATISTICAL SAFETY GUARANTEES | REGULATORY GOVERNANCE

### Latency Performance
**0.45 ms Fast-Path**
P99: 0.82 ms | 31,200 tx/s Throughput
Strict compliance with <10 ms SLA

### Workload Efficiency
**-99.51% Review Cut**
Only 49 of 10,000 transactions reviewed
85.7%--95.8% fewer false alerts vs rules

### Statistical Safety
**$\geq 99.0\%$ Coverage**
Finite-sample exchangeability bound
0.00% empty sets | Avg set size: 1.005

### Regulatory Governance
**SR 26-2 & OCC Ready**
Zero-hallucination graph evidence
Cryptographic SHA-256 Merkle audit trail